TEE-RPCH: Revocable Policy-based Chameleon Hashes via Trusted Execution Environments

ABSTRACT

This preview document instantiates the experimental section of the TEE-RPCH paper with the currently completed MNT224 real-SGX measurements. It is intended to let us inspect the final paper-style figure and table layout before the full dual-curve version is finished.

CCS CONCEPTS

• **Security and privacy** → **Cryptography**; *Trusted computing.*

KEYWORDS

trusted execution environment, chameleon hash, revocation, policy-based cryptography

1 INTRODUCTION

This preview focuses on integrating the completed MNT224 measurements into the paper template. The goal here is not to finalize the full manuscript, but to verify that the experiment figures, tables, and surrounding narrative can be rendered in the same style as the target paper draft.

2 PRELIMINARIES

The cryptographic and system preliminaries are omitted in this preview build. This file is included only so that the paper template can compile with the experiment section populated by measured data.

3 PCH2OA BASELINE

This preview build omits the full technical development of the baseline construction and keeps only the structure needed to display the measured experimental results.

4 TEE-RPCH CONSTRUCTION

This preview build omits the full construction details and keeps only the structure required to present the experimental figures and tables using the paper template.

5 EVALUATION

This preview section now reflects the corrected targeted measurements generated under hardware SGX for both MNT224 and A1. The table inputs are synchronized with the targeted-table pipeline so we can inspect the paper-style layout with the updated data.

5.1 Theory Comparison

5.2 Baseline Comparison

We compare the corrected split design CHET-2OA ABE-2OD with the no-cloud baseline CHET-OA ABE-OD. The server, TEE, and modifier columns are reported separately to match the actual implementation split.

5.3 Hash-check Benefit

We also isolate the user-side benefit of the final hash-check trick. The strict legacy comparison now measures a full re-encryption-based finalize path rather than the earlier lightweight AES-only check.

5.4 Revocation Comparison

For revocation, we compare only with server-assisted revocation designs, namely EHR-RPCH and TAR-PCH, together with our TEE-RPCH state-signing and TEE-check path.

Table 1: Theory comparison of representative revocable policy-based chameleon hash schemes.

Scheme	KeyGen	Hash	Adapt / Revocation Path
FB-PCH	$O\|S\|$	$O\|policy\|$	$O\|policy\|$
EHR-RPCH	$O\|S\| \log N$	$O\|policy\|$	KGC token + server ciphertext update
PNITCH	$O\|S\|$	$O\|policy\|$	authorization token + adapt
RPCH-XNM'21	$O\|S\| \log N$	$O\|policy\|$	$O\|policy\| \log N$
TAR-PCH	$O\|S\| \log N$	$O\|policy\|$	KGC trace + server update
TPCH	$O\|S\| \log N$	$O\|policy\|$	$O\|policy\| \log N$
TEE-RPCH	$O\|S\|$	$O\|policy\|$	server Transform1 + TEE Transform2/check + user hash-check

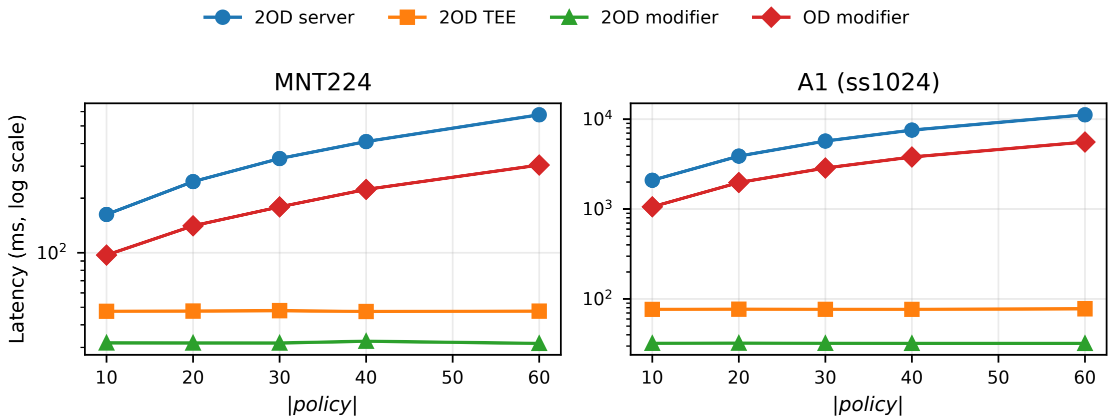

Figure 1: Adaptation latency split on MNT224 and A1.

Table 2: Adaptation latency across policy sizes (ms), $|\mathbb{S}| = 60$.

$\|policy\|$	CHET-2OA + ABE-2OD						CHET-OA + ABE-OD			
	Server		TEE		Modifier		TEE		Modifier	
	MNT	A1	MNT	A1	MNT	A1	MNT	A1	MNT	A1
10	162.5	2093.0	47.5	76.8	31.8	32.2	50.3	79.4	97.0	1060.3
20	246.6	3885.8	47.6	77.1	31.8	32.3	50.3	79.4	140.7	1974.8
30	330.5	5711.0	47.9	76.9	31.7	32.1	50.3	79.0	178.9	2863.2
40	410.0	7562.1	47.4	76.8	32.5	32.1	50.3	79.2	223.3	3797.6
60	576.2	11193.3	47.6	77.9	31.6	32.1	50.2	79.1	304.3	5572.4

Table 3: User-side finalization comparison (ms).

$\|policy\|$	Full Re-encryption Finalize		Re-encryption Check Only		Hash-check Finalize		Hash Check Only	
	MNT	A1	MNT	A1	MNT	A1	MNT	A1
10	350.7	6351.7	318.9	6319.6	31.8	32.2	20.2	20.2
20	1122.2	22604.7	1090.5	22572.4	31.8	32.3	20.3	20.2
30	2394.4	48654.4	2362.7	48622.3	31.7	32.1	20.2	20.2
40	4083.5	84556.3	4051.0	84524.3	32.5	32.1	20.6	20.2
60	8927.7	185295.0	8896.1	185263.0	31.6	32.1	20.2	20.2

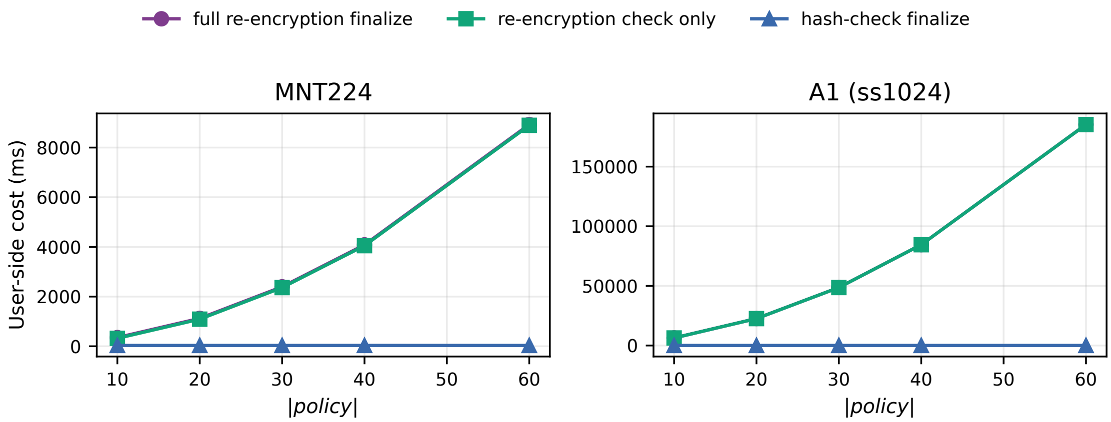

Figure 2: Full re-encryption finalization versus hash-check finalization.

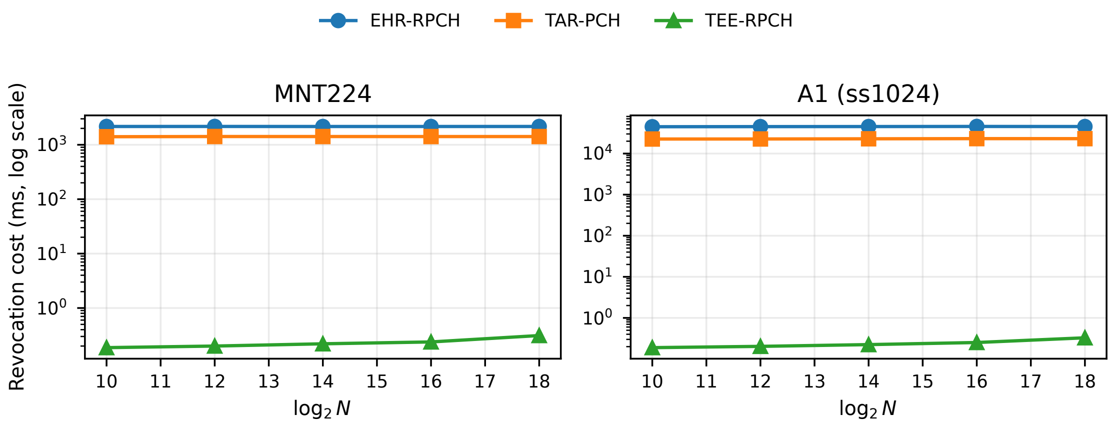

Figure 3: Revocation cost for one modifier on MNT224 and A1.

Table 4: Revocation cost for one revoked modifier (ms).

N	EHR-RPCH						TAR-PCH						TEE-RPCH					
	KGC		Server		Total		KGC		Server		Total		KGC		TEE Check		Total	
	MNT	A1	MNT	A1	MNT	A1	MNT	A1	MNT	A1	MNT	A1	MNT	A1	MNT	A1	MNT	A1
2^{10}	0.001	0.001	2168.870	44982.100	2168.870	44982.100	0.001	0.001	1406.650	22581.600	1406.650	22581.600	0.041	0.043	0.146	0.146	0.187	0.189
2^{12}	0.001	0.000	2172.430	45288.800	2172.430	45288.800	0.001	0.000	1418.850	22645.400	1418.850	22645.400	0.043	0.044	0.157	0.159	0.200	0.203
2^{14}	0.001	0.001	2167.500	45441.700	2167.500	45441.700	0.000	0.001	1417.520	22876.400	1417.520	22876.400	0.046	0.049	0.174	0.175	0.220	0.224
2^{16}	0.001	0.001	2168.220	45697.800	2168.220	45697.800	0.001	0.000	1416.690	23090.200	1416.690	23090.200	0.052	0.054	0.186	0.198	0.238	0.252
2^{18}	0.001	0.001	2170.410	45471.300	2170.410	45471.300	0.000	0.001	1416.860	23057.300	1416.860	23057.300	0.067	0.071	0.245	0.257	0.312	0.328

6 CONCLUSION

The MNT224 preview confirms that the current figure and table assets can be embedded into the target paper format. The full report still needs the corresponding A1 measurements before the final experimental section can be completed.

A OPEN SCIENCE

The artifacts needed to evaluate the experimental claims of this paper include the TEE-RPCH implementation, benchmark scripts, plotting scripts, configuration files, and raw timing logs. These artifacts will be provided to the program committee through an anonymous artifact package for double-blind review. No external user data or proprietary datasets are required to reproduce the reported cryptographic and systems measurements.

B ETHICAL CONSIDERATIONS

This work studies a cryptographic primitive and its TEE-assisted revocation mechanism. It does not involve human subjects, user data, or real-world vulnerability disclosure. The main security risk is misuse of redactable mechanisms to perform unauthorized history modification; our construction is designed to restrict rewriting to authorized modifiers and to support revocation of such privileges.

C ADDITIONAL MATERIAL

No additional appendix material is included in this preview build.